# SL Unit Quiz Solutions

CS 7641: Machine Learning

## 1 Question 1 - Decision Trees

## Dataset

| Example | Weather | SoilMoisture | Temperature | TimeAvailable | Garden |
|---|---|---|---|---|---|
| 1 | Sunny | Dry | Warm | Yes | Yes |
| 2 | Rainy | Wet | Cool | No | No |
| 3 | Overcast | Moist | Warm | Yes | Yes |
| 4 | Sunny | Dry | Hot | No | No |
| 5 | Rainy | Wet | Mild | Yes | No |
| 6 | Overcast | Moist | Cool | Yes | Yes |
| 7 | Sunny | Moist | Mild | Yes | Yes |

### Part 2. Decision Tree Applications (MCMA)

Answer the following questions based on the gardening dataset. Select all that apply.

**Question 1: Interpreting Leaf Nodes** You end up with a leaf node where all examples had the same label. What can you conclude?

A. This node will not be split further.

B. This node has zero entropy.

C. All features contributed equally to this node.

D. The path to this node may include multiple splits.

E. There must be at least two examples at this node.

F. ID3 assigns the most common label at this node.

**Answer:** A, B, D

**Question 2: Handling Ties in Information Gain** Suppose two attributes yield the same highest information gain. What options does ID3 have?

A. It will choose randomly between them.

B. It will prefer the one with fewer unique values.

C. It will halt tree construction due to ambiguity.

D. It may follow a fixed attribute order as a tiebreaker.

E. It will select the one that appears first in the dataset.

F. It splits on both attributes in parallel.

**Answer:** A, D, E

**Question 3: Decision Trees and Generalization** After training the tree on your gardening data, you test on new data. Which practices help generalization?

A. Prune branches that overfit training data.

B. Avoid splitting if the information gain is very low.

C. Add more attributes to better capture patterns.

D. Limit tree depth or minimum examples per leaf.

E. Include the label as an input to prevent underfitting.

F. Perform cross-validation to tune stopping criteria.

**Answer:** A, B, D, F

## Part 3. ID3 Calculation (with All Attributes)

a) **Entropy of the target variable Garden:**

- Yes: 4 examples (1, 3, 6, 7) - No: 3 examples (2, 4, 5)

$$H(Garden) = -\left(\frac{4}{7}\log_2\frac{4}{7} + \frac{3}{7}\log_2\frac{3}{7}\right) \approx -(0.571 \cdot -0.807 + 0.429 \cdot -1.222) = 0.985$$

b) **Information Gain for Weather**

- Sunny (1, 4, 7): 2 Yes, 1 No → Entropy = 0.918 - Rainy (2, 5): 0 Yes, 2 No → Entropy = 0 - Overcast (3, 6): 2 Yes, 0 No → Entropy = 0

$$H(Garden \mid Weather) = \frac{3}{7}(0.918) + \frac{2}{7}(0) + \frac{2}{7}(0) = 0.393$$

$$IG(Garden, Weather) = 0.985 - 0.393 = 0.592$$

c) **Information Gain for SoilMoisture**

- Dry (1, 4): 1 Yes, 1 No → Entropy = 1.0 - Wet (2, 5): 0 Yes, 2 No → Entropy = 0 - Moist (3, 6, 7): 3 Yes, 0 No → Entropy = 0

$$H(Garden \mid SoilMoisture) = \frac{2}{7}(1.0) + \frac{2}{7}(0) + \frac{3}{7}(0) = 0.286$$
$$IG(Garden, SoilMoisture) = 0.985 - 0.286 = 0.699$$

d) **Information Gain for TimeAvailable**

- Yes (1, 3, 5, 6, 7): 4 Yes, 1 No → Entropy =

$$-\left(\frac{4}{5}\log_2\frac{4}{5} + \frac{1}{5}\log_2\frac{1}{5}\right) \approx 0.722$$

- No (2, 4): 0 Yes, 2 No → Entropy = 0

$$H(Garden \mid TimeAvailable) = \frac{5}{7}(0.722) + \frac{2}{7}(0) = 0.516$$
$$IG(Garden, TimeAvailable) = 0.985 - 0.516 = 0.469$$

e) **Information Gain for Temperature**

- Warm (1, 3): 2 Yes, 0 No → Entropy = 0 - Cool (2, 6): 1 Yes, 1 No → Entropy = 1.0 - Hot (4): 0 Yes, 1 No → Entropy = 0 - Mild (5, 7): 1 Yes, 1 No → Entropy = 1.0

$$H(Garden \mid Temperature) = \frac{2}{7}(0) + \frac{2}{7}(1.0) + \frac{1}{7}(0) + \frac{2}{7}(1.0) = \frac{4}{7} = 0.571$$
$$IG(Garden, Temperature) = 0.985 - 0.571 = 0.414$$

f) **Which attribute will ID3 select at the root node?**

- SoilMoisture: $IG = 0.699$
- Weather: $IG = 0.592$
- TimeAvailable: $IG = 0.469$
- Temperature: $IG = 0.414$

**Answer:** ID3 will select **SoilMoisture** as the root node because it gives the highest information gain.

## 2 Question 2 - Regression & Classification

### Part 3. Regularization Effects (MCMA)

You are training a linear regression model on a dataset with many correlated features. You decide to apply L2 regularization (Ridge Regression).

Select all that apply:

A. L2 regularization penalizes large weights in the model.

B. L2 regularization will reduce training error.

C. The penalty term added is proportional to the square of each coefficient.

D. L2 regularization increases model complexity.

E. Increasing the regularization strength $\lambda$ typically increases bias.

F. L2 regularization helps mitigate multicollinearity.

**Answer:** A, C, E, F

### Part 5. Decision Boundaries (MCMA)

You visualize a classification model and observe decision boundaries between two classes.

Select all that apply:

A. A linear classifier produces straight-line boundaries in feature space.

B. Decision trees generate piecewise-linear boundaries.

C. A highly curved boundary suggests low variance.

D. Support vector machines with an RBF kernel can produce complex, nonlinear boundaries.

E. Logistic regression can only model linear boundaries unless features are transformed.

F. Neural networks always produce linear boundaries between classes.

**Answer:** A, B, D, E

### Part 6. Evaluation Metrics (MCMA)

You are evaluating a binary classifier on an imbalanced dataset (95% Class 0, 5% Class 1).

Select all that apply:

A. Accuracy may be misleading in this setting.

B. Precision and recall provide more insight than accuracy.

C. ROC AUC is a good metric for ranking classifier outputs.

D. The F1 score balances precision and recall.

E. A high accuracy guarantees high recall.

F. A perfect classifier would have an AUC of 0.5.

**Answer:** A, B, C, D

## Part 7. Choosing $k$ in k-NN Classification (MCMA)

You are tuning the value of $k$ in a $k$-Nearest Neighbors classifier for a binary classification task. Select all that apply:

A. A smaller $k$ tends to result in lower bias and higher variance.

B. As $k$ increases, the decision boundary becomes smoother.

C. Setting $k = 1$ always leads to the best generalization on noisy data.

D. Very large $k$ values can underfit the data.

E. Choosing an odd value for $k$ helps avoid ties in binary classification.

F. Cross-validation can help identify the optimal $k$.

**Answer:** A, B, D, E, F

## Part 4. Comparing Polynomial Regression Models with Error Metrics

You're predicting the number of hats produced based on daily labor hours. Two regression models are proposed:

- **Model A (Quadratic):** $\hat{y}_A = 2x^2 - 5x + 35$
- **Model B (Cubic):** $\hat{y}_B = -x^3 + 12x^2 - 37x + 50$

You test the models on the following validation data:

| Observation | Labor Hours $x$ | True Hats Produced $y$ |
|---|---|---|
| 1 | 5 | 50 |
| 2 | 6 | 65 |
| 3 | 7 | 60 |
| 4 | 8 | 80 |
| 5 | 9 | 95 |

a) Compute predicted values $\hat{y}$ for both models.

b) Compute the Mean Squared Error (MSE).

c) Compute the Mean Absolute Error (MAE).

d) Compute the L1 norm of the error vector.

e) Compute the L2 norm of the error vector.

f) Based on all metrics, which model performs better? Justify briefly.

**Instructor Key:**

**a) Predicted values**

| Obs | $x$ | $y$ | $\hat{y}_A = 2x^2 - 5x + 35$ | $\hat{y}_B = -x^3 + 12x^2 - 37x + 50$ |
|---|---|---|---|---|
| 1 | 5 | 50 | 60 | 40 |
| 2 | 6 | 65 | 77 | 44 |
| 3 | 7 | 60 | 98 | 36 |
| 4 | 8 | 80 | 123 | 10 |
| 5 | 9 | 95 | 152 | $-40$ |

**b) Mean Squared Error (MSE)**

$$\text{MSE}_A = \frac{(-10)^2 + (-12)^2 + (-38)^2 + (-43)^2 + (-57)^2}{5} = \frac{6786}{5} = \boxed{1357.2}$$

$$\text{MSE}_B = \frac{(-10)^2 + (-21)^2 + (-24)^2 + (-70)^2 + (-135)^2}{5} = \frac{24242}{5} = \boxed{4848.4}$$

**c) Mean Absolute Error (MAE)**

$$\text{MAE}_A = \frac{10 + 12 + 38 + 43 + 57}{5} = \boxed{32}$$

$$\text{MAE}_B = \frac{10 + 21 + 24 + 70 + 135}{5} = \boxed{52}$$

**d) L1 Norm**

$$\text{L1}_A = 10 + 12 + 38 + 43 + 57 = \boxed{160}$$

$$\text{L1}_B = 10 + 21 + 24 + 70 + 135 = \boxed{260}$$

**e) L2 Norm**

$$\text{L2}_A = \sqrt{6786} \approx \boxed{82.38}, \quad \text{L2}_B = \sqrt{24242} \approx \boxed{155.68}$$

**f) Conclusion:**
Model A (quadratic) still performs better than Model B (cubic) across all error metrics.
The updated values reinforce that Model B overfits and generalizes poorly to validation data.

# 3 Question 3 - Neural Netwoks and Gradient Descent

## Part 1. Learning Rate Strategies in Practice (MCMA)

You are tuning a neural network and experimenting with learning rate schedules.
Select all correct strategies:

A. Using a warm-up phase can stabilize training in early epochs.

B. Learning rate decay schedules are commonly used in modern deep learning.

C. Exponentially increasing the learning rate helps find local minima faster.

D. Cyclical learning rates periodically increase and decrease the learning rate.

E. Adaptive optimizers make fixed learning rate schedules unnecessary.

F. Gradient clipping is often paired with high learning rates to avoid instability.

**Answer:** A, B, D, F

## Part 2. Representing XOR with a Two-Layer MLP (MCMA)

Select all true statements about using a two-layer MLP to represent the XOR function:

A. A single-layer perceptron cannot solve XOR due to its non-linearity.

B. A two-layer MLP with ReLU activations can represent XOR.

C. At least one hidden layer is required to compute XOR using a feedforward neural network.

D. Any activation function can be used to solve XOR, even a linear one.

E. The hidden layer must introduce a non-linear transformation to separate XOR classes.

F. With a two-layer MLP, exactly two neurons in the hidden layer are always sufficient for XOR.

**Answer:** A, B, C, E

## Part 3. Training a Two-Layer MLP on XOR (MCMA)

You are training a two-layer MLP (1 hidden layer, 2 neurons, sigmoid activation) to solve the XOR problem.
Select all that are typically true:

A. Random initialization can lead to convergence to a suboptimal local minimum.

B. Gradient descent may struggle early if both hidden units are inactive.

C. Without non-linearity in the hidden layer, the network cannot learn XOR.

D. With sigmoid activations, vanishing gradients can slow down training.

E. XOR can be perfectly learned with a linear activation and enough training time.

F. Adding more hidden units can help but is not strictly necessary.

**Answer:** A, B, C, D, F

## Part 4. Two-Layer Network with ReLU (Modified)

You are working with a two-layer neural network to model a simple regression task. The structure and parameters are as follows:

- Input: $x = 2$
- First layer: weight $w_1 = 1.2$, activation: ReLU
- Second layer: weight $w_2 = 0.8$, activation: identity
- True label: $y = 4$
- Learning rate: $\eta = 0.1$
- Loss function: Squared error $L = \frac{1}{2}(y - \hat{y})^2$

**Tasks:**

a) Compute the forward pass and output $\hat{y}$

b) Compute the gradient $\frac{dL}{dw_2}$

c) Compute the gradient $\frac{dL}{dw_1}$

d) Perform one step of gradient descent to update both weights

## Solution

**a) Forward pass:**

$$z_1 = w_1 \cdot x = 1.2 \cdot 2 = 2.4$$
$$a_1 = \text{ReLU}(z_1) = \max(0, 2.4) = 2.4$$
$$\hat{y} = w_2 \cdot a_1 = 0.8 \cdot 2.4 = 1.92$$

**b) Gradient with respect to $w_2$:**

$$\frac{dL}{d\hat{y}} = \hat{y} - y = 1.92 - 4 = -2.08$$
$$\frac{dL}{dw_2} = \frac{dL}{d\hat{y}} \cdot \frac{d\hat{y}}{dw_2} = -2.08 \cdot a_1 = -2.08 \cdot 2.4 = -4.992$$

**c) Gradient with respect to $w_1$:**

$$\frac{dL}{dw_1} = \frac{dL}{d\hat{y}} \cdot \frac{d\hat{y}}{da_1} \cdot \frac{da_1}{dz_1} \cdot \frac{dz_1}{dw_1}$$
$$\frac{dL}{dw_1} = (-2.08) \cdot w_2 \cdot 1 \cdot x = -2.08 \cdot 0.8 \cdot 2 = -3.328$$

**d) Gradient descent updates:**

$$w_2^{\text{new}} = w_2 - \eta \cdot \frac{dL}{dw_2} = 0.8 - 0.1 \cdot (-4.992) = 0.8 + 0.4992 = \boxed{1.2992}$$
$$w_1^{\text{new}} = w_1 - \eta \cdot \frac{dL}{dw_1} = 1.2 - 0.1 \cdot (-3.328) = 1.2 + 0.3328 = \boxed{1.5328}$$

# 4 Question 4 - Computational Learning Theory (PAC Learning)

A hypothesis class $\mathcal{H}$ is said to be **PAC (Probably Approximately Correct)** learnable if there exists a learning algorithm that, for any $\epsilon, \delta > 0$, can find a hypothesis $h \in \mathcal{H}$ such that:

$$\Pr[\text{error}_D(h) > \epsilon] \leq \delta$$

with sample complexity:

$$m \geq \frac{1}{\epsilon}\left(\ln|\mathcal{H}| + \ln\left(\frac{1}{\delta}\right)\right)$$

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### Part 3. PAC Learning: Hypothesis Guarantees (MCMA)

Select all statements that are true about the guarantees provided by PAC learning:

A. PAC learning bounds apply to any arbitrary hypothesis, not just the learned one.

B. With probability at least $1 - \delta$, the learned hypothesis has error at most $\epsilon$.

C. PAC learning provides worst-case guarantees over data distributions.

D. PAC learning assumes the true concept is within the hypothesis class.

E. The PAC model requires that all hypotheses be deterministic.

F. PAC learning guarantees the hypothesis with the smallest training error.

**Answer:** B, C, D

### Part 4. PAC Learning and Sample Complexity (MCMA)

Select all statements that correctly describe how sample complexity behaves in PAC learning:

A. Sample complexity increases as the hypothesis space becomes more expressive.

B. Sample complexity is unaffected by the confidence parameter $\delta$.

C. Reducing the allowed generalization error $\epsilon$ increases sample requirements.

D. In PAC learning, sample complexity can be bounded without assumptions on the data distribution.

E. Infinite sample complexity implies the hypothesis class is not PAC learnable.

F. Consistent learners always achieve optimal sample complexity.

**Answer:** A, C, D, E

## Part 3. PAC Sample Complexity: Varying Parameters (Numerical)

Assume $|\mathcal{H}| = 10{,}000$. Compute the sample complexity under different accuracy and confidence settings:

a) $\epsilon = 0.10$, $\delta = 0.05$

b) $\epsilon = 0.05$, $\delta = 0.01$

c) $\epsilon = 0.01$, $\delta = 0.001$

Use the formula:

$$m \geq \frac{1}{\epsilon}\left(\ln|\mathcal{H}| + \ln\left(\frac{1}{\delta}\right)\right)$$

## PAC Sample Complexity Calculations

Let's compute the **sample complexity** $m$ for each scenario using the formula:

$$m \geq \frac{1}{\epsilon}\left(\ln|\mathcal{H}| + \ln\left(\frac{1}{\delta}\right)\right)$$

We are given:

$$|\mathcal{H}| = 10{,}000 \Rightarrow \ln|\mathcal{H}| = \ln(10^4) = 4\ln(10) \approx 4 \cdot 2.3026 \approx 9.2103$$

We'll now calculate each case:

**a)** $\epsilon = 0.10$**,** $\delta = 0.05$

$$\ln\left(\frac{1}{\delta}\right) = \ln\left(\frac{1}{0.05}\right) = \ln(20) \approx 2.9957$$

$$m \geq \frac{1}{0.10}(9.2103 + 2.9957) = 10 \cdot 12.206 \approx \boxed{122.06} \Rightarrow \boxed{123}$$

**b)** $\epsilon = 0.05$**,** $\delta = 0.01$

$$\ln\left(\frac{1}{\delta}\right) = \ln(100) \approx 4.6052$$

$$m \geq \frac{1}{0.05}(9.2103 + 4.6052) = 20 \cdot 13.8155 \approx \boxed{276.31} \Rightarrow \boxed{277}$$

**c)** $\epsilon = 0.01$**,** $\delta = 0.001$

$$\ln\left(\frac{1}{\delta}\right) = \ln(1000) \approx 6.9078$$

$$m \geq \frac{1}{0.01}(9.2103 + 6.9078) = 100 \cdot 16.1181 \approx \boxed{1611.81} \Rightarrow \boxed{1612}$$

**Final Answers**

- a) $m \geq \boxed{123}$
- b) $m \geq \boxed{277}$
- c) $m \geq \boxed{1612}$

## Part 6. VC Dimension Interpretation (MCMA)

Select all true statements about the role of VC dimension:

A. VC dimension measures the expressiveness of a hypothesis class.

B. A higher VC dimension always implies better generalization.

C. VC dimension corresponds to the largest number of points that can be shattered.

D. If a class has infinite VC-dim, it cannot be PAC learnable.

E. The VC-dim of linear classifiers in $R^d$ is $d + 1$.

F. Two classes with the same VC dimension have the same sample complexity.

**Answer:** A, C, D, E

## Part 4. VC Dimension of Geometric Classifiers in $R^{10}$ (Numerical)

For the following classifiers in $R^{10}$, determine the **exact** or **tightest known bound** on VC dimension:

a) VC dimension of all hyperplanes in $R^{10}$.

b) VC dimension of all 10-dimensional spheres (hyperspheres) with arbitrary centers and radii.

c) VC dimension of axis-aligned 10-dimensional boxes (hyperrectangles).

## Instructor Key: VC Dimension of Geometric Classifiers in $R^{10}$

a) **VC dimension of all hyperplanes in $R^{10}$:**

A hyperplane in $R^d$ has VC dimension $d + 1$, since it can shatter any set of $d + 1$ affinely independent points, but not $d + 2$.

$$\text{VC-dim} = \boxed{11}$$

b) **VC dimension of all 10-dimensional spheres (hyperspheres):**

A hypersphere in $R^{10}$ is defined by 10 coordinates for the center and 1 for the radius, giving 11 degrees of freedom. One might initially conjecture that this implies a VC dimension of 11, especially since this matches the VC dimension of hyperplanes. However, due to the nonlinear

nature of hypersphere boundaries, the class of such classifiers can shatter more complex point sets.

In fact, as shown in *High-Dimensional Statistics* by Martin Wainwright (see Figure 1), the VC dimension of spheres in $R^d$ is known to be exactly $d + 2$. For $d = 10$, this gives:

$$\text{VC-dim} = \boxed{12}$$

**Example 4.22** (Spheres in $\mathbb{R}^d$) Consider the sphere $S_{a,b} := \{x \in \mathbb{R}^d \mid \|x - a\|_2 \leq b\}$, where $(a, b) \in \mathbb{R}^d \times \mathbb{R}_+$ specify its center and radius, respectively, and let $\mathcal{S}^d_{\text{sphere}}$ denote the collection of all such spheres. If we define the function

$$f_{a,b}(x) := \|x\|_2^2 - 2\sum_{j=1}^{d} a_j x_j + \|a\|_2^2 - b^2,$$

then we have $S_{a,b} = \{x \in \mathbb{R}^d \mid f_{a,b}(x) \leq 0\}$, so that the sphere $S_{a,b}$ is a subgraph of the function $f_{a,b}$.

In order to leverage Proposition 4.20, we first define a feature map $\phi\colon \mathbb{R}^d \to \mathbb{R}^{d+2}$ via $\phi(x) := (1, x_1, \ldots, x_d, \|x\|_2^2)$, and then consider functions of the form

$$g_c(x) := \langle c, \phi(x) \rangle \qquad \text{where } c \in \mathbb{R}^{d+2}.$$

The family of functions $\{g_c, c \in \mathbb{R}^{d+1}\}$ is a vector space of dimension $d + 2$, and it contains the function class $\{f_{a,b}, (a, b) \in \mathbb{R}^d \times \mathbb{R}_+\}$. Consequently, by applying Proposition 4.20 to this larger vector space, we conclude that $\nu(\mathcal{S}^d_{\text{sphere}}) \leq d + 2$. This bound is adequate for many purposes, but is not sharp: a more careful analysis shows that the VC dimension of spheres in $\mathbb{R}^d$ is actually $d + 1$. See Exercise 4.13 for an in-depth exploration of the case $d = 2$. ♣

Figure 1: Shattering bound for hyperspheres in $R^d$ (from Wainwright, 2019).

*Note:* Some may be tempted to argue for a VC dimension of 11 based on degrees of freedom or by taking the radius to infinity (approaching hyperplanes), but this misses the richness of the hypothesis class. The tightest known result confirms a VC dimension of $d + 2$. However, since this is outside the scope of this course, both 11 and 12 will be accepted as answers.

c) **VC dimension of axis-aligned 10-dimensional boxes (hyperrectangles):**

An axis-aligned box in $R^{10}$ is defined by two thresholds per dimension (lower and upper), so the VC dimension is exactly $2d$.

$$\text{VC-dim} = \boxed{20}$$